Cybersecurity Fundamentals - Practical Assessment

Total Marks: 100 Marks **Assessment Type:** Practical + Scenario-Based

SECTION 1: NETWORKING FUNDAMENTALS

Task 1: Office Network Design Case Study

Scenario: Designing a network addressing and allocation plan for a four-floor corporate office building using assigned private space `192.168.10.0/24` with 20% capacity expansion for each department.

Q1. Device Requirement Calculations (+20% Expansion)

- **Ground Floor (Reception & Admin):** 20 × 1.20 = **24 devices**
- **First Floor (HR & Finance):** 35 × 1.20 = **42 devices**
- **Second Floor (Development Team):** 60 × 1.20 = **72 devices**
- **Third Floor (SOC & IT Operations):** 45 × 1.20 = **54 devices**
- **Server Room:** 12 × 1.20 = 14.4 → **15 devices**

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Q2. Subnetting Calculations

Applying Variable Length Subnet Masking (VLSM) ordered by host capacity demand:

- **Development Team (72 hosts):** Requires block size of 128 (2^7). Prefix: `/25` (Mask: `255.255.255.128`). Network: `192.168.10.0/25`. Range: `192.168.10.1` – `192.168.10.126`. Broadcast: `192.168.10.127`. Usable Hosts: 126.
- **SOC & IT Operations (54 hosts):** Requires block size of 64 (2^6). Prefix: `/26` (Mask: `255.255.255.192`). Network: `192.168.10.128/26`. Range: `192.168.10.129` – `192.168.10.190`. Broadcast: `192.168.10.191`. Usable Hosts: 62.
- **HR & Finance (42 hosts):** Requires block size of 64 (2^6). Prefix: `/26` (Mask: `255.255.255.192`). Network: `192.168.10.192/26`. Range: `192.168.10.193` – `192.168.10.254`. Broadcast: `192.168.10.255`. Usable Hosts: 62.

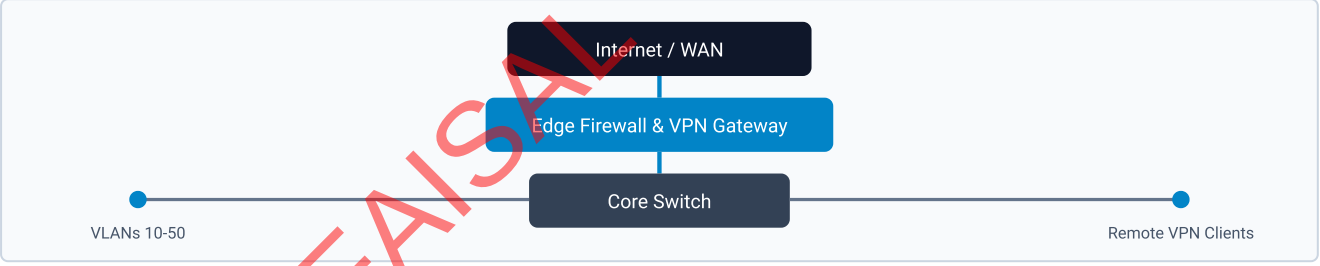
Q3. IP Address Allocation Table

Department	Network / CIDR	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address	Hosts
Development Team	192.168.10.0/25	255.255.255.128	192.168.10.1	192.168.10.126	192.168.10.127	126
SOC & IT Operations	192.168.10.128/26	255.255.255.192	192.168.10.129	192.168.10.190	192.168.10.191	62
HR & Finance	192.168.10.192/26	255.255.255.192	192.168.10.193	192.168.10.254	192.168.10.255	62
Reception & Admin	*Exhausted*	*N/A*	*N/A*	*N/A*	*N/A*	0
Server Room	*Exhausted*	*N/A*	*N/A*	*N/A*	*N/A*	0

Q4. Network Capacity Analysis

- **Usable IP Count:** A standard `/24` subnet provides 2^8 - 2 = **254 usable IP addresses**.
- **Fit Assessment:** No, all 5 departments cannot fit simultaneously within a single `/24` block. Although total required hosts (207) is less than 254, VLSM block alignment constraints (128 + 64 + 64 + 32 + 32 = 320 space) exceed the 254 limit.
- **Recommended Plan:** Expand to a `/23` network (e.g., `192.168.10.0/23`) offering 510 usable hosts or adopt Class B `10.10.0.0/16` private address space.
- **Future Growth Rationale:** Planning for growth prevents IP address exhaustion, service downtime, and manual reconfiguration of static infrastructure when new devices are deployed.

Q5. Network Architecture Diagram



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Q6. Security Recommendations

- **VLAN Segmentation & ACLs:** Isolate each department into separate Virtual Local Area Networks (VLANs 10–50) with Access Control Lists to restrict unauthorized lateral movement.
- **Stateful Server Firewall Rules:** Protect the server subnet by allowing access strictly on specific operational ports (e.g., SSH/443) from authorized IT/SOC management IPs.
- **MFA-Enforced Remote VPN:** Deploy TLS/IPsec VPN access integrated with Multi-Factor Authentication (MFA) to securely authenticate remote workers prior to network granting.
- **802.1X Port Security & WPA3 Enterprise:** Require RADIUS-based 802.1X authentication across switch ports and wireless APs to block unauthorized rogue physical connections.

Task 2: Networking Concepts and Troubleshooting

> **Q1. Private vs. Public IP:** Private IPs (e.g., `10.0.0.0/8`, `192.168.0.0/16`) are non-routable on the global internet and reserved for local networks. Public IPs are globally unique and internet-routable.

Q2. Subnetting Purpose: Subnetting divides large networks into smaller segments to limit broadcast domains, improve routing performance, reduce congestion, and apply granular security policies.

Q3. Default Gateway Role: A default gateway (e.g., `192.168.1.1`) serves as the routing exit node for local hosts (e.g., `192.168.1.25`) when transmitting packets targeted outside their immediate local subnet boundary.

Q4. DNS Resolution Failures: If access works by IP but fails by domain name, the issue stems from DNS failures—caused by misconfigured client DNS resolvers or DNS cache corruption. Investigate using `nslookup` or `dig`.

Q5. DHCP Functions: Dynamic Host Configuration Protocol automates IP, gateway, and DNS configuration. If DHCP is unreachable, hosts assign an APIPA address (`169.254.x.x`), losing external network connectivity.

Q6. TCP vs. UDP: TCP is a connection-oriented, reliable protocol guaranteed by handshakes (e.g., HTTPS). UDP is a connectionless, fast protocol without delivery guarantees (e.g., DNS, live video streaming).

Q7. VPN Security: A VPN creates an encrypted tunnel for remote access. Poorly configured VPNs without MFA risk allowing compromised remote hosts to act as entry points for threat actors.

Q8. Troubleshooting Workflow: Check physical link status → verify IP/gateway config → ping local gateway → ping public IP (8.8.8.8) → verify DNS resolution → run traceroute to locate path drops.

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Task 3: Practical Networking Commands

```
student@linux-lab:~$ ip addr show eth0 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq
_codel state UP inet 192.168.1.50/24 brd 192.168.1.255 scope global eth0 student@linux-lab:~$ ip ro
ute default via 192.168.1.1 dev eth0 proto dhcp src 192.168.1.50 student@linux-lab:~$ ping -c 2 19
2.168.1.1 64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.421 ms 2 packets transmitted, 2 recei
ved, 0% packet loss
```

SECTION 2: LINUX FUNDAMENTALS

Task 4: Linux Filesystem Navigation and Commands

```
student@linux-lab:~$ pwd && whoami && hostname && uname -r /home/student student security-workstati
on 5.15.0-88-generic student@linux-lab:~$ mkdir -p "October Assessment/Networking" "October Assessm
ent/Linux" "October Assessment/Reports" student@linux-lab:~$ touch "October Assessment/Networking/s
ubnetting.txt" "October Assessment/Linux/commands.txt" student@linux-lab:~$ echo "Networking fundam
entals and VLSM notes" > "October Assessment/Linux/commands.txt" student@linux-lab:~$ echo "Linux p
ermission and ownership controls" >> "October Assessment/Linux/commands.txt"
```

Filesystem Structure Explanation

- `/home`: Stores non-root personal user profiles and documents.
- `/etc`: Houses system-wide operational configuration files (e.g., `/etc/passwd`).
- `/var`: Stores variable runtime data like system logs (`/var/log`) and mail spools.
- `/usr`: Houses system user utilities, libraries, and binaries.
- `/bin`: Contains essential core binary commands required for system startup (e.g., `ls`, `bash`).
- `/root`: Dedicated root administrator home profile directory.

> **Path Absolute vs. Relative:** An absolute path starts from the system root directory `/` (e.g., `/home/student/October Assessment`). A relative path starts from the current working directory (e.g., `./Linux/commands.txt`).

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Task 5: Linux Permissions and File Management

```
student@linux-lab:~$ touch network_report.txt && chmod 640 network_report.txt student@linux-lab:~$
ls -l network_report.txt -rw-r----- 1 student staff 0 Oct 16 09:12 network_report.txt student@linux
-lab:~$ chmod 600 network_report.txt student@linux-lab:~$ ls -l network_report.txt -rw------- 1 stu
dent staff 0 Oct 16 09:14 network_report.txt
```

Permission 600 (-rw-------): Grants Read and Write access exclusively to the file owner. All access is revoked for group members and other users on the system.

Task 6: Linux Mini Challenge - IT Investigation

```
student@linux-lab:~$ mkdir -p "Linux Investigation/logs" "Linux Investigation/backup" student@linux
-lab:~$ cat << 'EOF' > "Linux Investigation/logs/application.log" 2026-10-16 09:10:12 INFO Service
started 2026-10-16 09:12:45 ERROR Failed to connect to database 2026-10-16 09:13:10 WARNING Retryin
g connection 2026-10-16 09:14:02 ERROR Database connection timeout EOF student@linux-lab:~$ grep "E
RROR" "Linux Investigation/logs/application.log" 2026-10-16 09:12:45 ERROR Failed to connect to dat
abase 2026-10-16 09:14:02 ERROR Database connection timeout student@linux-lab:~$ grep -c "ERROR" "L
inux Investigation/logs/application.log" 2 student@linux-lab:~$ cp "Linux Investigation/logs/applic
ation.log" "Linux Investigation/backup/"
```

Investigation Questions & Findings

- **Error Search Command:** Executed `grep "ERROR" application.log` to filter lines.
- **Error Entry Count:** Exactly 2 matching database connection failure error lines identified.
- **Log Backup Value:** Preserves original forensic evidence intact for analysis prior to log rotation or tampering.
- **Missing vs Permission Denied:** A missing file returns `No such file or directory`, whereas restricted file access returns `Permission denied`.

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